

Code Listing

1 The Basic Verbatim Environment

```
for i in range(len(X)):
    # iterate through columns
    for j in range(len(X[0])):
        result[j][i] = X[i][j]
```

This is an inline example of `\verbatim` in $\text{\LaTeX}$.

An enhanced verbatim text using fancyvrb package.

2 The listings Package

2.1 List Inline

This is an inline Example of listings package `for i in range(len(X)):` – and that was the code.

2.2 Normal Blocks of Code: `lstlisting`

```
def my_function():
    print("Hello from a function")
```

2.3 Code From Source File

```
public class Greetings {
    public static void main( String[] args ) {
        System.out.println( "Hello world!" );
    }
}
```

2.4 Package Control Options

```
1      # Question No. 1
2      def sum_geometric_series(a,r,n):
3          term = 0
4          # Start Numbering at One for Right Exponent
5          for i in range(1, n+1):
6              term += a * pow(r, i) # Accumulating
7          series_sum = a + term
8          return series_sum
9
10     # Example
11     print("The Result is:",sum_geometric_series(2,3,4))
12
```

Listing 1: Geometric Series Sum.

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```
13
14     # Question No. 2
15     def approximate_e(n):
16         e = pow((1+1/2), n )
17         return e
18
19     # Example
20     print(approximate_e(2))
```

Listing 2: Approximate Euler Number.

2.5 Listing Specific Range of Lines

```
1     for k in range(N + 1):
2         x = a + (k * dx)
3         summand = f(x)
4         if (k != 0) and (k != N):
5             summand *= (2 + (2 * (k - 2)))
6         value += summand
```

3 The minted Package

```
1     # Question No. 8:  $F = [F_x, F_y]$ 
2     import numpy as np
3     from math import *
4     from numpy import array
5
6     def force_vec(F1, F2, theta1, theta2):
7         Rx = F1 * sin(radians(theta1)) + F2 *
8             ↪ cos(radians(theta2))
9         Ry = F1 * cos(radians(theta1)) + F2 *
10            ↪ sin(radians(theta2))
11
12         R = [Rx, Ry]
13         return R
14
15     # Example
16
17     F1 = 250
18     F2 = 450
19
20     theta1 = 30
21     theta2 = 35
22
23     print("Result:", force_vec(F1, F2, theta1, theta2))
```

```
1 import library
```

This is an Inline Code `Print("Hello World !!")` Offered by minted Package.